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INTERNSHIP REPORT

on

A TinyML-Based Method for Real-Time State of Charge Estimation Using TCN on STM32

Submitted by

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Jan – May 2026

Carried under the guidance of

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CERTIFICATE

This is to certify that the report entitled “**A TinyML-Based Method for Real-Time State of Charge Estimation Using TCN on STM32**” is a bonafide work carried out by **Manish C (PES1UG22EC152)** in partial fulfillment of the requirements for the completion of 8th semester internship work in the Program of Study, B.Tech in Electronics and Communication Engineering, under the rules and regulations of PES University, Bengaluru during the period Jan – May 2026.

It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report.

Signature with Date
Guide / Faculty / Mentor

DECLARATION

I hereby declare that the work entitled “**A TinyML-Based Method for Real-Time State of Charge Estimation Using TCN on STM32**” has been carried out by me under the guidance of **Guide Name**, PES University.

I, **Manish C**, declare that the work entitled “*A TinyML-Based Method for Real-Time State of Charge Estimation Using TCN on STM32*” is an original work carried out under the guidance of **Vadiraja Acharya**. This report is being submitted in partial fulfillment of the requirements for completion of 8th Semester course work in the Program of Study, B.Tech in Electronics and Communication Engineering.

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Abstract

This report presents a TinyML-based approach for real-time State of Charge (SoC) estimation of lithium-ion batteries using a Temporal Convolutional Network (TCN) deployed on an STM32 microcontroller. Accurate SoC estimation is essential in Battery Management Systems (BMS) as it directly influences battery safety, performance, and lifespan. It enables efficient energy utilization, prevents overcharging and deep discharge, and ensures reliable operation in applications such as electric vehicles and portable electronics. However, direct measurement of SoC is not feasible, and traditional estimation methods suffer from cumulative error and sensitivity to operating conditions.

Conventional techniques such as Coulomb counting and model-based approaches like Kalman filtering rely on simplified assumptions or accurate battery models, which are difficult to maintain under real-world conditions. These methods often fail to capture the nonlinear and time-dependent behavior of batteries under varying load and temperature. Data-driven approaches address these limitations by learning complex relationships directly from measurable parameters.

In this work, voltage, current, and temperature are measured using sensors interfaced with an STM32 microcontroller through ADC. A custom dataset is generated through controlled discharge experiments, capturing realistic battery behavior. Since direct SoC measurement is not available, a data-driven formulation is used to approximate SoC values based on voltage, with corrections incorporating current and temperature variations. The data is further processed using normalization and temporal windowing for model training.

A Temporal Convolutional Network (TCN) is used to model the sequential nature of battery data. Unlike recurrent architectures, TCN employs causal and dilated convolutions to efficiently capture long-range dependencies while supporting parallel computation, making it suitable for embedded deployment. The trained model is optimized using 8-bit quantization and deployed on STM32 using TensorFlow Lite and X-CUBE-AI.

The system performs real-time inference on streaming sensor data, providing continuous SoC estimation with low latency and memory usage. Experimental results demonstrate a Mean Absolute Error (MAE) of approximately 2.25% with stable performance under varying conditions. This work highlights the effectiveness of integrating TinyML with embedded systems for practical and efficient battery monitoring applications.

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1 Introduction

Lithium-ion (Li-ion) batteries have become the dominant energy storage solution in modern electronic systems due to their high energy density, long cycle life, and efficiency. They are extensively used in applications such as electric vehicles, portable electronics, renewable energy storage systems, and industrial backup systems. In these applications, reliable monitoring and management of battery performance are essential to ensure safety, efficiency, and longevity. Battery Management Systems (BMS) are responsible for monitoring critical parameters and maintaining optimal operating conditions.

One of the most important parameters in a BMS is the State of Charge (SoC), which represents the remaining capacity of a battery as a percentage of its total capacity. Accurate estimation of SoC enables better energy utilization, improves system reliability, prevents overcharging and deep discharge, and enhances overall battery lifespan. However, SoC cannot be directly measured using physical sensors and must be estimated indirectly using measurable signals such as voltage, current, and temperature.

Traditional methods for SoC estimation include Coulomb counting and model-based approaches such as Kalman filtering. Coulomb counting estimates SoC by integrating current over time, but it suffers from cumulative error due to sensor noise, drift, and inaccuracies in initial conditions. Model-based approaches rely on equivalent circuit models or electrochemical models of batteries, which require precise parameter identification. These models are often difficult to maintain under real-world operating conditions where battery behavior varies with temperature, load, and aging effects.

To overcome these limitations, data-driven approaches based on machine learning have gained significant attention. These methods can learn the nonlinear relationship between measurable parameters and battery state directly from data, eliminating the need for explicit modeling. Conventional machine learning techniques such as Support Vector Regression (SVR), Random Forest (RF), and Gaussian Process Regression (GPR) have demonstrated improved performance compared to traditional methods. However, these approaches often struggle to capture temporal dependencies in battery behavior, which are critical for accurate SoC estimation.

Deep learning methods, including Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks, have been widely explored for SoC estimation due to their ability to extract spatial and temporal features from sequential data. Hybrid architectures combining CNN and LSTM have further improved prediction accuracy. Despite their effectiveness, these models are computationally intensive and require significant memory and processing resources, making them unsuitable for deployment on resource-constrained embedded platforms such as microcontrollers.

Recent advancements in TinyML have enabled the deployment of machine learning models directly on low-power embedded devices. TinyML focuses on optimizing models to achieve low memory footprint, reduced computational complexity, and real-time inference capabilities. This enables intelligent decision-making directly on edge devices without

the need for cloud connectivity, thereby reducing latency, improving reliability, and enhancing system autonomy. Such capabilities are particularly important in battery management systems, where continuous monitoring and real-time response are essential.

Temporal Convolutional Networks (TCNs) have emerged as a promising alternative to recurrent architectures for modeling sequential data. Unlike recurrent networks, TCNs use causal and dilated convolutions to capture long-range temporal dependencies efficiently. The use of dilation allows the receptive field to expand without increasing network depth significantly, enabling efficient learning of temporal patterns. Additionally, TCNs support parallel computation, which reduces inference time and makes them suitable for real-time embedded applications.

In this work, a TinyML-based approach for real-time SoC estimation is proposed using a Temporal Convolutional Network deployed on an STM32-NucleoL432KC microcontroller. Unlike many existing studies that rely on offline datasets collected under controlled laboratory conditions, this work utilizes real sensor data obtained from a lithium-ion battery operating under practical conditions. The dataset includes measurements of voltage, current, and temperature, capturing the dynamic behavior of the battery under varying loads and environmental conditions.

A custom data acquisition setup is developed to collect sensor data during controlled discharge experiments. Since direct SoC measurement is not available, a data-driven formulation is used to generate target labels based on voltage, with additional corrections incorporating current and temperature variations. The collected data is preprocessed and organized into temporal sequences using a sliding window approach, enabling effective training of the TCN model.

The trained model is optimized using 8-bit quantization and deployed on the STM32 microcontroller using TensorFlow Lite and X-CUBE-AI. This ensures efficient execution with minimal memory usage while maintaining acceptable accuracy. The system performs real-time inference on streaming sensor data, providing continuous estimation of the battery's State of Charge.

Experimental results demonstrate that the proposed approach achieves a Mean Absolute Error (MAE) of approximately 2.25% with low inference latency. The model is also observed to be robust to sensor noise and variations in operating conditions, highlighting its practical applicability in real-world battery management systems.

The main contributions of this work can be summarized as follows:

- Development of a real-time TinyML-based SoC estimation system using a Temporal Convolutional Network.
- Implementation of a complete pipeline including sensor data acquisition, preprocessing, model training, and embedded deployment.
- Utilization of real-world sensor data instead of standardized offline datasets.

- Deployment of a quantized model on an STM32 microcontroller for efficient on-device inference.
- Demonstration of accurate and low-latency SoC estimation under practical operating conditions.

This work bridges the gap between high-accuracy data-driven SoC estimation techniques and their practical deployment on resource-constrained embedded systems, demonstrating the potential of TinyML for next-generation intelligent battery management solutions.

2 Literature Survey

Accurate estimation of battery State of Charge (SoC) has been an active research area due to its importance in battery management systems. Traditional methods such as Coulomb counting and Kalman filtering have been widely used; however, they suffer from cumulative error, sensitivity to noise, and dependency on accurate battery models.

To overcome these limitations, data-driven approaches based on machine learning have gained attention. Early works employed classical machine learning techniques such as Support Vector Machines (SVM), k-Nearest Neighbors (KNN), and Random Forests. These methods improved estimation accuracy compared to traditional techniques but struggled to capture temporal dependencies in battery behavior.

With the advancement of deep learning, models such as Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks have been widely explored for SoC estimation. CNN-based approaches focus on feature extraction from sensor data, while LSTM models are capable of learning temporal dependencies. Hybrid models combining CNN and LSTM have demonstrated improved performance by leveraging both spatial and temporal features.

Despite their high accuracy, deep learning models are computationally intensive and require significant memory resources, making them unsuitable for deployment on microcontrollers and other resource-constrained embedded systems.

Recent research has focused on TinyML-based approaches, which aim to deploy lightweight machine learning models on embedded hardware. These approaches enable real-time inference with low power consumption and minimal latency. However, achieving a balance between accuracy and computational efficiency remains a challenge.

Temporal Convolutional Networks (TCNs) have emerged as an efficient alternative for sequence modeling. Unlike recurrent networks, TCNs use causal and dilated convolutions to capture long-range temporal dependencies while supporting parallel computation. This makes them suitable for embedded applications where computational efficiency is critical.

In this work, a TinyML-based TCN model is proposed for real-time SoC estimation using an STM32 microcontroller. Unlike many existing studies that rely on offline datasets,

the proposed system utilizes real sensor data collected under practical operating conditions, enabling accurate and robust SoC estimation in real-world scenarios.

Paper	Platform	Inputs	Method	Application	Limitations	Accuracy	
CNN-Based SoC Estimation [1]	GPU/PC	Voltage, Current	CNN	EV Batteries	High computational cost, not suitable for embedded systems	High	
LSTM-Based Model [10]	GPU/PC	Time-series data	LSTM	Energy Storage	Sequential processing increases latency	High	
CNN-LSTM Hybrid [5, 11]	GPU/PC	Multi-parameter	CNN + LSTM	EV Systems	Complex architecture, high memory usage	Very High	
Kalman Filter Method [2, 12]	Embedded	Voltage, Current	Kalman Filter	BMS	Requires accurate model, sensitive to noise	Medium	
SVR-Based Estimation [9]	PC	Voltage	SVR	Battery Monitoring	Poor temporal dependency handling	Medium	
TinyML-Based Approach [15]	Microcontroller	Voltage, Current	TinyML	Embedded BMS	Trade-off between accuracy and model size	Moderate	
Proposed Method	TCN	STM32	Voltage, Current, Temp	TCN (TinyML)	Real-time BMS	–	~97–98%

Table 1: Comparison of Existing SoC Estimation Methods

3 Software Implementation

The software component of the proposed system plays a critical role in enabling data acquisition, preprocessing, model training, and real-time inference on embedded hardware. The complete software workflow is divided into two main stages: model development and embedded deployment.

The model development phase was carried out using Python in the Google Colab environment, which provides access to high-performance computational resources and pre-installed machine learning libraries. The primary framework used for model design and training is TensorFlow, along with its Keras API for building deep learning architectures. Data handling and preprocessing were performed using NumPy and Pandas, while visualization of results such as discharge curves and prediction performance was done using Matplotlib.

The dataset used for training was generated experimentally using a lithium-ion battery under controlled discharge conditions. The recorded parameters include voltage, current, and temperature, which were preprocessed through normalization to improve model convergence. A sliding window approach was applied to convert the sequential data into time-series input suitable for the Temporal Convolutional Network (TCN). This allowed the model to capture temporal dependencies in battery behavior.

The TCN model was implemented using a specialized Keras-compatible TCN library. The architecture consists of causal and dilated convolution layers, enabling efficient modeling of long-range temporal patterns while maintaining low computational complexity. The model was trained using the Adam optimizer with Mean Squared Error (MSE) as the loss

function. Performance evaluation was carried out using Mean Absolute Error (MAE), along with additional metrics such as accuracy and F1-score based on discretized SoC levels.

Once the model was trained and validated, it was converted into a lightweight format suitable for embedded deployment. TensorFlow Lite was used to convert the trained model into a compact representation, followed by 8-bit quantization to reduce memory usage and improve inference speed. This step is essential for deploying machine learning models on resource-constrained microcontrollers.

For embedded implementation, the STM32CubeIDE was used as the primary development environment. The X-CUBE-AI expansion package was utilized to integrate the TensorFlow Lite model into the STM32 microcontroller. This tool automatically converts the model into optimized C code and generates the necessary inference functions.

The microcontroller software is responsible for acquiring sensor data through the Analog-to-Digital Converter (ADC), performing preprocessing operations such as scaling and buffering, and feeding the processed data into the deployed TCN model. The predicted State of Charge (SoC) is then transmitted via UART for display or monitoring purposes.

Overall, the software framework ensures seamless integration between machine learning model development and embedded system deployment. The use of efficient libraries and optimization techniques enables real-time SoC estimation with minimal computational overhead, making the system suitable for practical battery management applications.

4 System Overview and Hardware Setup

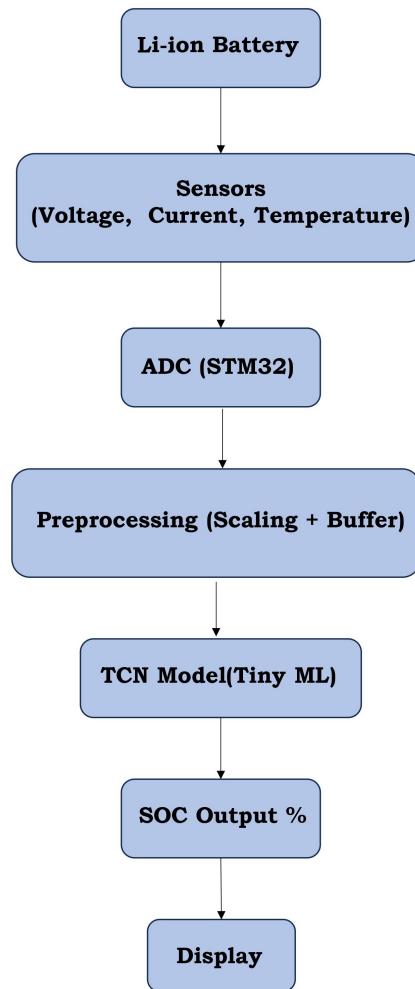


Figure 4.1: Overall workflow of the proposed TinyML-based SoC estimation system

The overall workflow of the proposed system is illustrated in Fig. 4.1. The process begins with a lithium-ion battery whose parameters are continuously monitored using sensors that measure voltage, current, and temperature. These analog signals are converted into digital values using the ADC of the STM32 microcontroller.

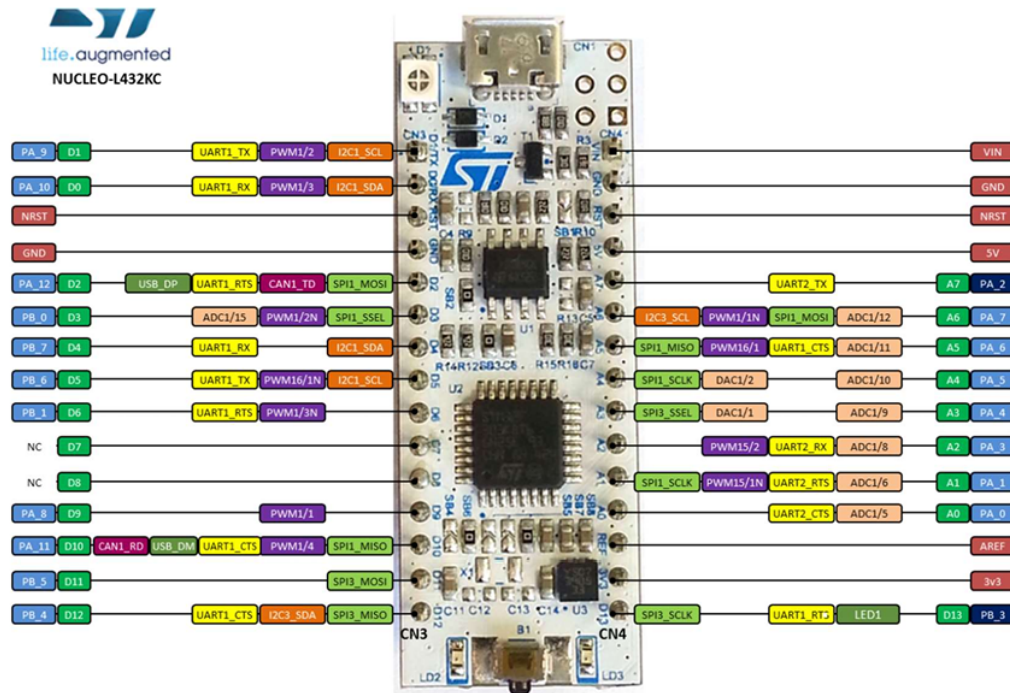
The acquired data is then preprocessed through scaling and buffering to ensure it is suitable for model input. The processed data is fed into a Temporal Convolutional Network (TCN) model deployed on the microcontroller using TinyML techniques. The model performs real-time inference to estimate the State of Charge (SoC), which is then displayed or transmitted for monitoring purposes.

4.1 STM32L432KC Microcontroller

The STM32L432KC microcontroller serves as the core processing unit of the proposed system. It is based on the ARM Cortex-M4 architecture and is designed for low-power embedded applications while providing sufficient computational capability for real-time inference tasks. The board includes integrated peripherals such as Analog-to-Digital Converters (ADC), communication interfaces (UART, I2C, SPI), and debugging support via ST-LINK.

In this system, the STM32 microcontroller is responsible for acquiring analog signals from sensors, performing preprocessing operations, and executing the deployed TinyML model for State of Charge (SoC) estimation. The ADC module converts analog voltage signals from the sensors into digital values, which are then scaled and buffered before being fed into the model.

The microcontroller also handles communication tasks, such as transmitting the estimated SoC values through UART for monitoring or display. Its low power consumption and real-time processing capability make it suitable for battery management applications.



The scaled voltage is fed into one of the ADC channels of the STM32 microcontroller. This simple and reliable method ensures accurate voltage measurement while protecting the microcontroller from overvoltage conditions.

4.3 ACS712 Current Sensor

The ACS712 current sensor is used to measure the current flowing through the battery. It operates based on the Hall-effect principle and provides an analog output voltage proportional to the current.

The sensor output is centered around a reference voltage (typically 2.5 V), allowing it to measure both charging and discharging currents. The output signal is further conditioned using a voltage divider before being connected to the ADC input of the STM32 microcontroller.

Current measurement is essential for improving SoC estimation accuracy, as battery voltage alone is insufficient under varying load conditions.

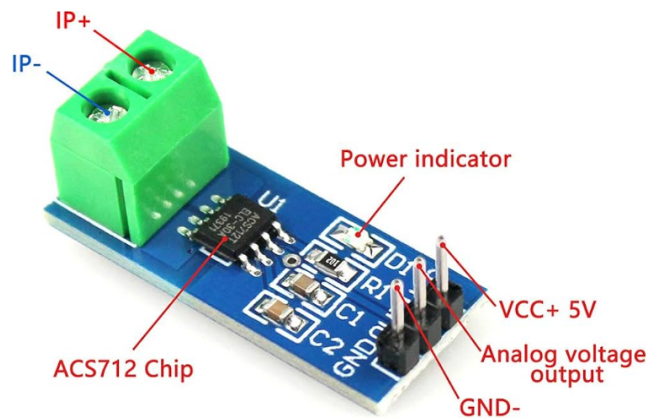


Figure 4.3: ACS712 Current Sensor

4.4 NTC Thermistor for Temperature Measurement

Temperature is measured using a Negative Temperature Coefficient (NTC) thermistor arranged in a voltage divider configuration. The resistance of the thermistor decreases with increasing temperature, resulting in a measurable change in output voltage.

This voltage is read by the ADC and converted into temperature values through calibration. Temperature monitoring is important for battery safety and enhances the robustness of SoC estimation under varying environmental conditions.



Figure 4.4: NTC Thermistor

4.5 Setup

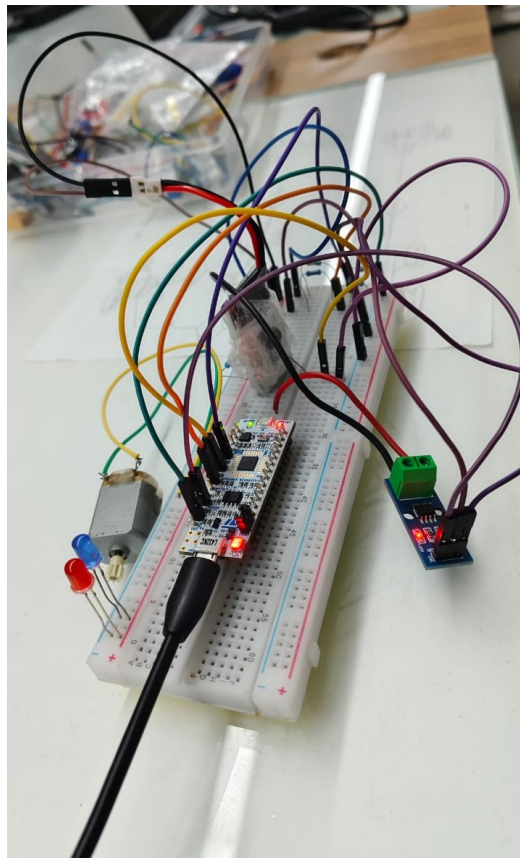


Figure 4.5: Experimental hardware prototype of the proposed TinyML-based SoC estimation system

Fig. 4.5 shows the experimental hardware setup used to validate the proposed system. The STM32 microcontroller is interfaced with voltage, current, and temperature sensing cir-

circuits using a breadboard-based implementation. The ACS712 current sensor module and voltage divider network are used for current and voltage measurement, respectively, while temperature is measured using an NTC thermistor.

The acquired analog signals are fed into the ADC channels of the microcontroller, where preprocessing and TinyML-based inference are performed in real time. The setup demonstrates the practical feasibility of deploying the proposed SoC estimation model on resource-constrained embedded hardware.

The hardware setup of the proposed system is designed to acquire battery parameters in real time using simple and reliable analog circuits interfaced with the STM32L432KC microcontroller. A 3.7 V lithium-ion battery is used as the primary source, and its voltage, current, and temperature are continuously monitored using dedicated sensing circuits.

Battery voltage is measured using a resistive voltage divider consisting of two 10 k Ω resistors. The divided voltage is connected to one of the ADC input pins of the STM32 microcontroller Current ADC Pin(e.g., A5 (PA6 / ADC12_IN11)). This ensures that the input voltage remains within the safe operating range of the ADC.

Current measurement is performed using the ACS712 Hall-effect current sensor. The sensor is connected in series with the battery load, and its analog output is fed to another ADC channel Current ADC Pin: (e.g., A1 (PA1 / ADC12_IN6)). The output signal is centered around a reference voltage, allowing detection of current variations during discharge.

Temperature is measured using a 10 k Ω NTC thermistor configured in a voltage divider circuit. The output voltage varies with temperature and is connected to a separate ADC input pin (e.g., Current ADC Pin: A3 (PA4 / ADC12_IN9)). This enables monitoring of thermal conditions, which are important for accurate SoC estimation.

All sensor outputs are interfaced with the STM32 ADC channels, and the microcontroller is powered using a stable 3.3 V supply. Proper grounding is ensured across all components to reduce measurement noise and improve signal stability.

The ADC is configured with an appropriate sampling rate to capture real-time variations in battery parameters (e.g., **[Sampling Rate: 247.5 samples/sec]**). The acquired data is scaled and buffered before being used for inference. This setup provides a reliable hardware interface for collecting multi-parameter data required for the TinyML-based SoC estimation system.

4.6 System Integration

The overall hardware setup consists of a lithium-ion battery connected to the sensing circuits for voltage, current, and temperature measurement. All sensor outputs are interfaced with the STM32 microcontroller through ADC channels.

The acquired data is processed in real time, where scaling and buffering operations are applied before passing the data to the deployed TCN-based TinyML model. The model predicts the State of Charge (SoC), which is then transmitted via UART for monitoring purposes.

The system is designed to operate under real-time conditions with minimal latency, making it suitable for embedded battery management applications. The use of simple analog circuits combined with efficient embedded processing ensures a balance between accuracy, cost, and computational efficiency.

5 Implementation

5.1 Data Acquisition and Preprocessing

A custom dataset was generated using a 3.7 V lithium-ion battery under controlled discharge conditions. The battery was connected to practical loads such as a DC motor and resistive elements to emulate real-world usage scenarios where current demand varies dynamically. This ensured that the collected data captures realistic battery behavior, including voltage sag, recovery effects, and nonlinear discharge characteristics.

Voltage, current, and temperature were continuously measured using dedicated sensing circuits and interfaced with the STM32L432KC microcontroller through its internal Analog-to-Digital Converter (ADC). The sampling process was configured to capture sufficient temporal resolution while maintaining stability in readings. The acquired signals inherently contain noise due to switching loads, sensor limitations, and ADC quantization effects.

To improve data quality, unstable operating regions—specifically voltage values below 2.45 V and above 3.7 V—were excluded, as these regions exhibited high fluctuations and unreliable measurements. This restriction ensures that the dataset represents a stable and meaningful operating range for model training.

Preprocessing was performed using Python in the Google Colab environment. The raw sensor values were normalized to a fixed range to prevent dominance of any single feature during training and to improve convergence speed. Additionally, outliers and abrupt spikes were smoothed using simple filtering techniques.

To capture temporal behavior, a sliding window approach was applied. Sequential data was segmented into overlapping windows of fixed length, where each sample consists of 10 consecutive timesteps with three features (voltage, current, and temperature). This results in an input tensor of shape (10, 3), enabling the model to learn temporal dependencies rather than relying on instantaneous values.

5.2 SoC Label Generation

Since direct measurement of State of Charge (SoC) is not available without specialized equipment, a data-driven approximation was used to generate target labels for supervised learning. Initially, SoC was estimated using a linear mapping of voltage within the stable operating range, where higher voltage corresponds to higher charge level.

However, voltage alone does not fully represent battery state under varying load conditions due to internal resistance and transient effects. To address this, correction factors based on current and temperature variations were incorporated into the estimation process. This allows the generated SoC labels to reflect real operating conditions more accurately.

To ensure temporal consistency, smoothing techniques such as exponential averaging were applied to reduce abrupt changes in SoC values caused by noise or sudden load variations. This step is particularly important for training stability, as inconsistent labels can negatively affect model learning.

Although the generated SoC values do not represent exact electrochemical states, they provide a consistent and practical approximation suitable for training machine learning models in embedded applications.

5.3 Temporal Convolutional Network (TCN) Model

A Temporal Convolutional Network (TCN) was selected for modeling the sequential nature of battery data due to its ability to efficiently capture temporal dependencies while maintaining low computational complexity. Unlike recurrent neural networks, TCN processes sequences using convolutional operations, allowing parallel computation and faster inference.

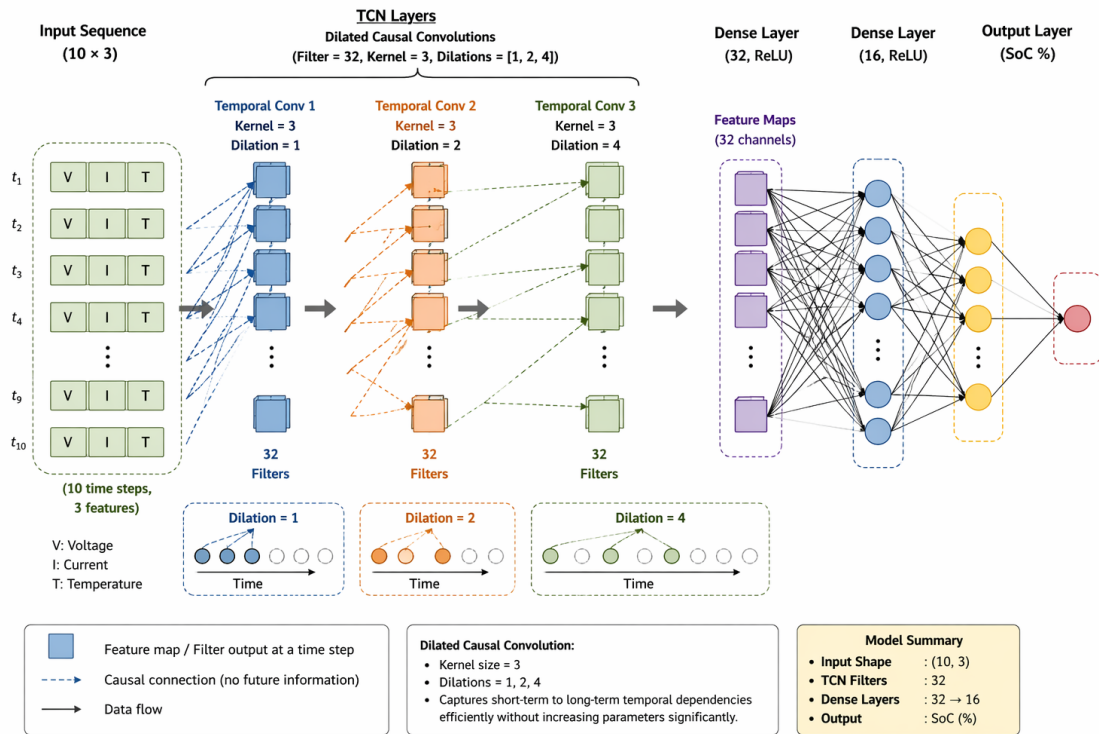


Figure 5.1: Temporal Convolutional Network (TCN) architecture for SoC estimation

As shown in the Fig. 5.1, The model uses causal convolutions to ensure that predictions at any time step depend only on past and present inputs, which is essential for real-time systems. Dilated convolutions with dilation rates of [1, 2, 4] were employed to expand

the receptive field exponentially, enabling the model to capture long-range dependencies without increasing network depth significantly.

The TCN layer consists of 32 filters with a kernel size of 3, allowing the extraction of temporal features from input sequences. The output of the TCN is passed through fully connected layers with ReLU activation functions, which introduce non-linearity and refine feature representation.

This architecture provides a balance between model complexity and performance, making it suitable for deployment on resource-constrained embedded platforms.

5.4 Model Training and Evaluation

The model was implemented and trained using TensorFlow and Keras in the Google Colab environment. The dataset was divided into training and validation subsets to evaluate generalization performance and prevent overfitting.

The Adam optimizer was used due to its adaptive learning rate and efficient convergence properties. Mean Squared Error (MSE) was selected as the loss function, as it penalizes larger errors more significantly and is suitable for regression tasks.

Model performance was evaluated using Mean Absolute Error (MAE), which provides an intuitive measure of prediction accuracy in percentage terms. Additionally, classification-based metrics such as accuracy and F1-score were computed by discretizing SoC values into categories (e.g., low, medium, high). This provides further insight into model reliability across different charge levels.

The training process demonstrated stable convergence, and the model was able to closely track the actual SoC trends observed during battery discharge experiments.

5.5 Model Optimization and Deployment

To enable deployment on the STM32 microcontroller, the trained model was converted into TensorFlow Lite format. Post-training quantization was applied to reduce the precision of weights and activations to 8-bit integers. This significantly reduces memory usage and improves inference speed, which is critical for embedded systems.

The quantized model was then deployed using the X-CUBE-AI framework within STM32CubeIDE. This tool converts the TensorFlow Lite model into optimized C code, enabling seamless integration into the embedded application.

Care was taken to ensure that the model size fits within the available memory constraints of the microcontroller while maintaining acceptable prediction accuracy.

5.6 Real-Time Embedded Implementation

During real-time operation, the STM32L432KC microcontroller continuously acquires sensor data through ADC channels (e.g., A1 (PA1 / ADC12_IN6), A2 (PA2 / ADC12_IN7),

A4 (PA4 / ADC12_IN9)). The analog signals are converted into digital values, scaled appropriately, and stored in a buffer.

A rolling buffer mechanism is used to maintain a fixed-length sequence of recent measurements, which forms the input to the TCN model. This ensures that temporal context is preserved during inference.

The buffered data is passed to the deployed model, which performs inference and outputs the predicted SoC value. The prediction is computed in real time with minimal latency and computational overhead.

The estimated SoC is transmitted via UART for display or logging purposes. The system operates efficiently under embedded constraints, demonstrating the feasibility of deploying machine learning models on microcontrollers using TinyML techniques.

Overall, the implementation presents a complete pipeline from data acquisition to real-time inference, highlighting the integration of hardware and software components for intelligent battery monitoring.

6 Results and Discussion

This section presents the performance evaluation of the proposed TinyML-based State of Charge (SoC) estimation system. The evaluation includes analysis of prediction accuracy, temporal behavior, and model performance under real-world operating conditions. The results are based on the custom dataset collected using a lithium-ion battery under controlled discharge conditions.

6.1 Battery Discharge Characteristics

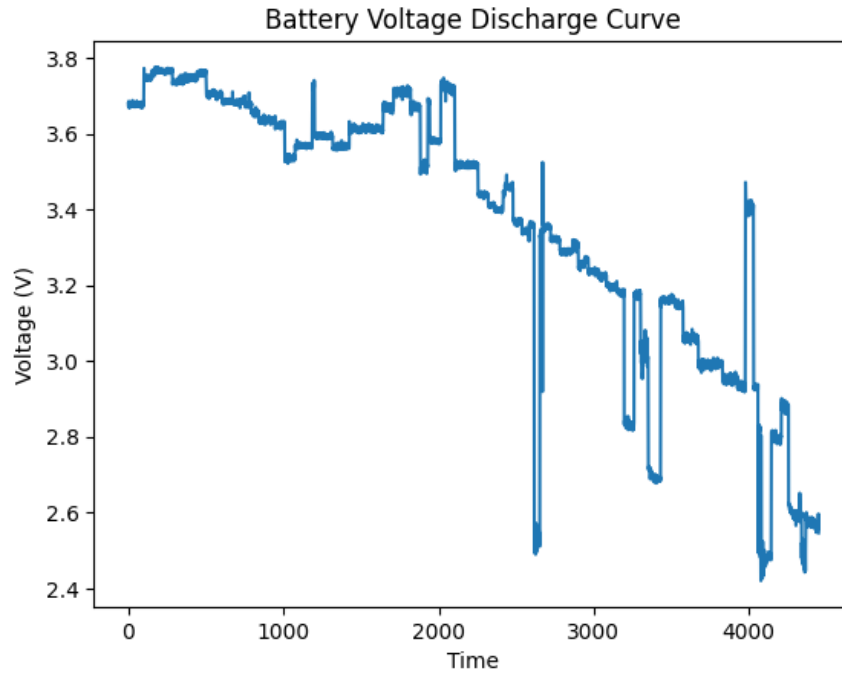


Figure 6.1: Battery voltage discharge curve under dynamic load conditions

As shown in Fig. 6.1, the battery discharge curve represents the variation of voltage with respect to time during the discharge process. As expected, the voltage decreases gradually from approximately 3.7 V to 2.45 V, indicating the depletion of battery charge.

The curve exhibits nonlinear behavior, where the voltage drop is relatively slow in the mid-region and more rapid near the lower voltage range. Additionally, sudden dips and spikes can be observed due to dynamic load conditions and internal resistance effects. These fluctuations highlight the limitations of voltage-only SoC estimation methods, as voltage does not consistently reflect the actual charge state under varying loads.

The inclusion of current and temperature measurements in the proposed system allows the model to capture these variations and improve estimation accuracy.

6.2 Actual vs Predicted SoC Analysis

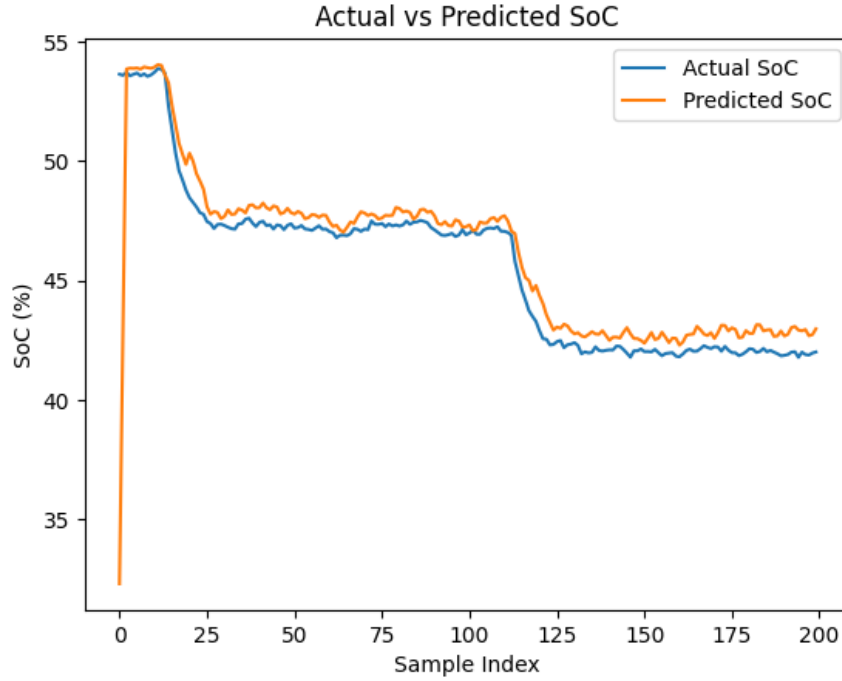


Figure 6.2: Comparison of actual and predicted SoC values

Fig. 6.2, the performance of the TCN model is evaluated by comparing the predicted SoC values with the actual SoC values obtained from the dataset. The comparison plot demonstrates that the predicted SoC closely follows the actual SoC trend throughout the discharge cycle.

The model successfully captures both gradual changes and sudden variations in SoC, indicating its ability to learn temporal dependencies in battery behavior. Minor deviations can be observed during rapid transitions, which are primarily caused by abrupt load changes and sensor noise. However, these errors remain small and do not significantly impact overall performance.

The smooth nature of the predicted curve also indicates that the model generalizes well and avoids overfitting to noise in the data.

6.3 Error Analysis

To quantitatively evaluate model performance, Mean Absolute Error (MAE) was computed. The model achieved an MAE of approximately 2.25%, indicating high prediction accuracy.

This low error demonstrates that the proposed approach is capable of estimating SoC with high reliability. Compared to traditional voltage-based methods, which can exhibit significant errors under dynamic conditions, the proposed model provides more stable and consistent predictions.

The error distribution was observed to be relatively uniform across different SoC levels, indicating that the model performs consistently across the entire operating range.

6.4 Classification-Based Evaluation

In addition to regression metrics, the model performance was evaluated using classification-based metrics by discretizing SoC values into categories such as low, medium, and high.

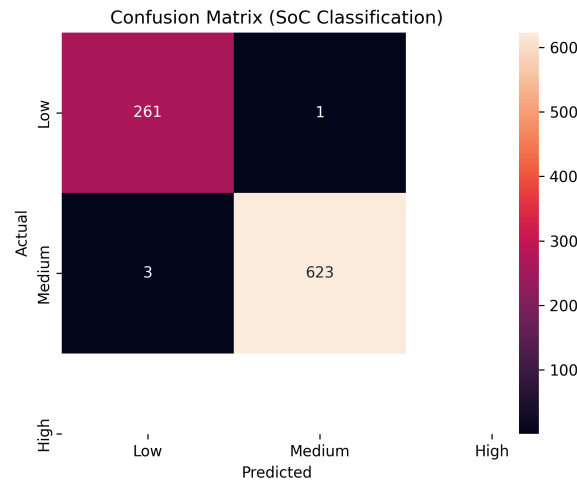


Figure 6.3: Confusion matrix for SoC classification

The confusion matrix in Fig. 6.3 shows that the majority of predictions fall along the diagonal, indicating correct classification. Only a small number of misclassifications were observed, primarily between adjacent classes, which is expected due to the continuous nature of SoC.

The model achieved an accuracy of approximately 99.5% and an F1-score of approximately 0.995, demonstrating excellent classification performance. These results indicate that the model is highly reliable in identifying battery charge levels in practical applications.

6.5 Real-Time Embedded Performance

The deployed model was tested on the STM32L432KC microcontroller to evaluate its real-time performance. The system successfully performed inference on streaming sensor data with low latency and minimal computational overhead.

The use of 8-bit quantization significantly reduced memory usage while maintaining prediction accuracy. The model fits within the memory constraints of the microcontroller and executes efficiently without requiring external processing.

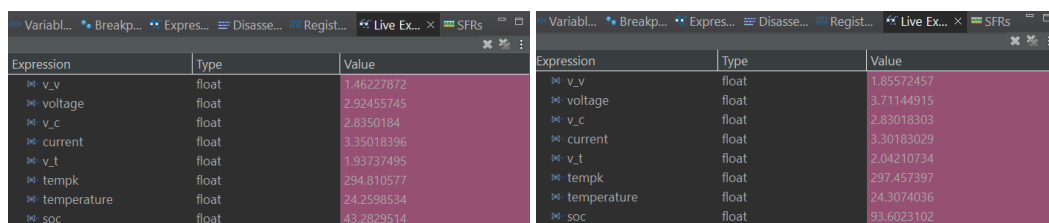


Figure 6.4: Real-time STM32 debug outputs showing sensor values and predicted SoC

Fig. 6.4 shows real-time variable monitoring during embedded execution. The system acquires voltage, current, and temperature values, processes them, and generates corresponding SoC predictions. The variation in SoC values reflects real-time battery conditions, demonstrating successful on-device inference using the deployed TinyML model and thus the real-time predictions were stable and consistent, demonstrating the feasibility of deploying machine learning models on resource-constrained embedded systems.

6.6 Discussion

The results clearly demonstrate the effectiveness of the proposed TinyML-based SoC estimation system. By incorporating multiple parameters such as voltage, current, and temperature, the system overcomes the limitations of traditional methods that rely solely on voltage.

The TCN model effectively captures temporal dependencies, enabling accurate prediction even under dynamic load conditions. The combination of high accuracy, low latency, and efficient embedded implementation makes the proposed approach suitable for real-world battery management systems.

Overall, the system provides a practical and scalable solution for real-time SoC estimation, bridging the gap between machine learning models and embedded deployment.

7 Conclusion

This work presents a complete and practical implementation of a TinyML-based State of Charge (SoC) estimation system using a Temporal Convolutional Network (TCN) deployed on an STM32 microcontroller. The proposed system successfully integrates data acquisition, preprocessing, model development, optimization, and real-time embedded deployment into a unified pipeline, demonstrating the feasibility of applying machine learning techniques in resource-constrained environments.

One of the key strengths of this work lies in the use of multi-parameter sensing, where voltage, current, and temperature are jointly utilized for SoC estimation. Unlike conventional voltage-based methods, which are highly sensitive to load variations and internal resistance effects, the proposed approach provides a more reliable and stable estimation by incorporating multiple physical parameters. This significantly improves prediction accuracy under dynamic operating conditions.

The use of a Temporal Convolutional Network enables efficient modeling of sequential battery behavior. By leveraging causal and dilated convolutions, the model captures long-range temporal dependencies while maintaining a relatively simple and computationally efficient architecture. This makes the model particularly suitable for embedded systems where both memory and processing power are limited.

A major contribution of this work is the successful deployment of the trained model on the STM32L432KC microcontroller using TinyML techniques. Through 8-bit quantization

and optimization using TensorFlow Lite and X-CUBE-AI, the model achieves a significantly reduced memory footprint while maintaining high prediction accuracy. The model operates within the limited RAM and Flash constraints of the microcontroller, demonstrating efficient utilization of available resources. In fact, the implementation highlights that accurate machine learning inference can be achieved even when memory is minimally utilized, making the solution highly scalable for low-cost embedded systems.

The system demonstrates low inference latency and stable real-time performance, which are critical requirements for battery management systems. The ability to perform on-device inference eliminates the need for cloud connectivity, reducing communication overhead, improving response time, and enhancing system reliability.

The proposed approach has wide applicability across multiple domains. It can be effectively used in electric vehicles for accurate battery monitoring and improved energy management. In portable electronics such as smartphones, laptops, and wearable devices, the system can enhance battery life prediction and user experience. It is also suitable for renewable energy storage systems, where accurate SoC estimation is essential for efficient power management. Additionally, the system can be applied in uninterruptible power supplies (UPS), industrial cordless tools, robotics, drones, and medical devices, where reliable battery monitoring is critical for safety and performance.

Furthermore, the lightweight nature of the model makes it highly suitable for large-scale deployment in Internet of Things (IoT) applications. Since the model requires minimal memory and computational resources, it can be integrated into low-power edge devices without significantly increasing system cost or complexity. This opens up possibilities for intelligent battery monitoring in distributed systems and smart energy networks.

Overall, this work successfully bridges the gap between high-accuracy machine learning models and their practical deployment on embedded hardware. It demonstrates that TinyML can be effectively used to implement intelligent, real-time battery monitoring systems with high accuracy, low latency, and efficient resource utilization. The proposed system provides a scalable and cost-effective solution for next-generation battery management applications.

Future work can focus on extending the model to handle battery aging effects, incorporating larger and more diverse datasets, and exploring adaptive learning techniques for further improving accuracy and robustness under varying real-world conditions.

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